Universitatea Babeş-Bolyai, Facultatea de Matematică şi Informatică

Secţia: Informatică engleză, Curs: Dynamical Systems

Laboratory 4.

Exercise 1.

Let $\lambda_1 = -2$, $\lambda_2 = -1 + i$, $\lambda_3 = -1 - i$,

$$u_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \; u_2 = \begin{pmatrix} 1+i \\ 1 \\ 0 \end{pmatrix}, \; u_3 = \begin{pmatrix} 1-i \\ 1 \\ 0 \end{pmatrix}, \; P = \begin{pmatrix} u_1 & u_2 & u_3 \end{pmatrix},$$

$$A = \begin{pmatrix} 0 & -2 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \qquad D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}.$$

(a) Compute $Au_1 - \lambda_1 u_1$, $Au_1 - \lambda_2 u_1$, $Au_2 - \lambda_2 u_2$, $Au_2 - \lambda_3 u_2$, $Au_3 - \lambda_3 u_3$. Deduce that u_k is an eigenvector corresponding to the eigenvalue λ_k, for each $k \in \{1, 2, 3\}$.

(b) Compute $A - PDP^{-1}$. Deduce that A is diagonalizable over $\mathbb{C}$.

(c) Compute e^{tD} and e^{tA}.

The matrix e^{tD} has complex entries with nonzero imaginary part because D has complex entries with nonzero imaginary part.

The matrix A has real entries and we know that, by definition, e^{tA} has also real entries.

(d) Compute the limit as $t \to \infty$ for each entry of e^{tA}.

(e) True/False:

"Each solution of the differential system $X' = AX$ verifies $\lim_{t\to\infty} X(t) = 0_3$."

Here 0_3 is the null column vector in $\mathbb{R}^3$.

Exercise 2.

(a) Find a 4×4 real matrix A with the eigenvalues $2, 2, -1, 0$ which is not diagonal but it is diagonalizable over $\mathbb{R}$.

Hint: First find an invertible 4×4 real matrix P (there are many!), then introduce the diagonal matrix D with $2, 2, -1, 0$ on the main diagonal. Take $A := PDP^{-1}$. From this relation we deduce that A is similar to the diagonal matrix D, thus, by definition, it is diagonalizable.

(b) Find the determinant and the characteristic polynomial of A from (a). Find the eigenvectors of A. Find the Jordan form of A. If it is invertible, find the inverse of A.

Exercise 3.

Let $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & -2 & 0 \end{pmatrix}$.

(a) Find the eigenvectors and the Jordan form. Are A and B diagonalizable?

(b) Find e^{tA} and e^{tB}.

Exercise 4.

We consider the first order nonlinear autonomous differential equation

$$x' = 1 - x^2.$$

For each $\eta \in \mathbb{R}$ denote by $\varphi(t, \eta)$ the unique solution of the IVP

$$x' = 1 - x^2, \quad x(0) = \eta.$$

(a) Solve the algebraic equation $1 - x^2 = 0$. Find $\varphi(t, -1)$ and $\varphi(t, 1)$. Plot their graphs in your notebook and write their images denoted by γ_{-1} and, respectively, γ_1.

(b) Find $\varphi(t, 0)$. Use **convert(convert(tanh(t),exp),exp)** to obtain the expression

$$\varphi(t, 0) = \frac{e^{2t} - 1}{e^{2t} + 1}, \quad t \in \mathbb{R}.$$

Find $\lim_{t \to \infty} \varphi(t, 0)$ and $\lim_{t \to -\infty} \varphi(t, 0)$.

Plot its graph on the interval $\mathbb{R}$. Is this function strictly increasing?

Plot its graph in your notebook and write its image denoted by γ_0.

(c) Find $\varphi(t, -2)$. Use **convert(convert(tanh(t-arctanh(2)),exp),exp)** to obtain the expression

$$\varphi(t, -2) = \frac{e^{2t} + 3}{e^{2t} - 3}, \quad t \in (-\infty, \ln \sqrt{3}).$$

Find $\lim_{t \to -\infty} \varphi(t, -2)$ and $\lim_{t \to \ln \sqrt{3}} \varphi(t, -2)$.

Plot its graph on the interval $(-\infty, \ln \sqrt{3})$. Is this function strictly decreasing?

Plot its graph in your notebook and write its image denoted by γ_{-2}.

(d) Find $\varphi(t, 2)$. Obtain the expression

$$\varphi(t, 2) = \frac{3e^{2t} + 1}{3e^{2t} - 1}, \quad t \in (-\ln \sqrt{3}, \infty).$$

Find $\lim_{t \to -\ln \sqrt{3}} \varphi(t, 2)$ and $\lim_{t \to \infty} \varphi(t, 2)$.

Plot its graph on the interval $(-\ln \sqrt{3}, \infty)$. Is this function strictly decreasing?

Plot its graph in your notebook and write its image denoted by γ_2.